

Figure 1: The 'android-debug.apk' file copied to the phone



Figure 2: 'Install blocked' warning message



Figure 3: Click on 'OK', to allow the torch Cordova app to be installed on our phone

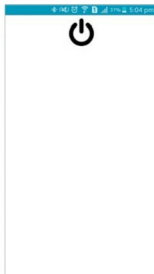


Figure 4: The torch app opened on Android mobile phone

- **initialize():** This is a default application constructor. It makes a call to **bindEvent()** when the application starts up.
- **bindEvent():** This is a bind event listener that binds events such as **load**, **deviceready**, **offline**, etc, which are required on start up. In the above code, this function is listening for the 'deviceready' event. Once the device is ready, it makes a call to **onDeviceReady()**.
- **onDeviceReady():** This is a 'deviceready' event handler; in this function, we make an explicit call to **app.receiveEvent()**, which specifies the device is ready. We have added the plugin code to turn on the flashlight of the mobile's camera in this function. We have added a click event listener to an element whose ID is 'torch', which means that when we click

that element, the **window.plugins.flashlight.toggle()** function is called. In simple words, it toggles the flashlight on/off, when we click the element.

- **receivedEvent():** This function is used to handle the **deviceready** event.

Building the app

We are done with the coding part; now let us build the .apk file of the app so that we can install it on our Android mobile phone.

Run the following command to build the project for the Android platform:

```
C:\project\torchapp>cordova build android
```

This command generates a .apk file at **C:\project\torchapp\platforms\android\build\outputs\apk** location in our project directory with the name 'android-debug.apk'.



Note: You can use the **cordova build** command to build the app for multiple target platforms such as iOS, Windows Phone, etc, by suitably replacing the keyword **android** in the above command statement.

Installing and testing the app

Now copy or transfer the **android-debug.apk** file to your phone via USB cable or Bluetooth, as shown in Figure 1, and install it (while installing, you may receive an 'Install blocked' message, as shown in Figure 2. So click **Settings**, check the 'Unknown Sources' option and select 'OK', as shown in Figure 3). This will now give you access permission of the app; next, click on 'Install'.

After installing the app, open it as shown in Figure 4. Since this is a torch application, we have created a white background for the app, so the user can use the bright screen light as well. When you open the app, you will see a black power button image at the top-centre. Click on it to toggle the mobile camera flashlight on/off.



Note: You can also refer to and download this project's source code from my GitHub repository. Visit <https://github.com/aniketkudale/torch-app/>.



References

- [1] <https://cordova.apache.org/>
- [2] https://en.wikipedia.org/wiki/Apache_Cordova
- [3] <https://cordova.apache.org/docs/en/latest/guide/overview/>

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